

Kevin Kattakayam

B.Tech Computer Science and Engineering (Artificial Intelligence & Machine Learning)

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PROFESSIONAL SUMMARY

Final-year B.Tech CSE (AI & ML) student working across applied generative AI and backend systems engineering: multi-agent LLM pipelines with LangGraph and CrewAI, RAG over real document corpora, self-supervised video models in PyTorch, and a Go/Rust streaming observability pipeline on Kafka, ClickHouse, Helm, and Terraform. IEEE-published, completed Amazon ML Summer School 2026, and holder of ten professional certifications across Microsoft, Snowflake, Salesforce, Oracle, Redis, and Google.

EDUCATION

Karunya Institute of Technology and Sciences

2023 – 2027 (Expected)

B.Tech, Computer Science and Engineering (AI & ML) – CGPA: 8.39/10

Coimbatore, India

Viswajyothi CMI Public School, Angamaly – Secondary, CBSE (2021): 92.2%; Higher Secondary, CBSE (2023): 86%

PROFESSIONAL EXPERIENCE

TheAiSignal – Full Stack AI Developer Intern

May 2026 – Jul 2026

- Engineered AI-powered features spanning a React frontend, FastAPI backend, and REST API layer for production applications.
- Maintained deployment pipelines and release workflows across multiple projects using GitHub and Cloudflare.

Karunya Computer Technology Centre – Generative AI Trainee

Dec 2024 – Present

- Build LLM-powered applications with a team across multiple generative AI projects using LangChain, ChromaDB, and FAISS.
- Apply retrieval-augmented generation, prompt engineering, and OCR-based extraction to document processing workflows.

YoungMinds Technology Solutions Pvt Ltd – Generative AI Intern

May 2025 – Jul 2025

- Developed generative AI applications and automation tools in Python against LLM and REST APIs.

Karunya Innovation and Design Studio (KIDS) – Generative AI Intern

May 2025 – Jun 2025

- Built a multi-agent patent drafting and verification system in CrewAI as the internship's hands-on project, following training across generative AI concepts (detailed under Projects: PatentDoc Co-Pilot).

Octanet Services Pvt Ltd – Web Development Intern

May 2024 – Jun 2024

- Built responsive web applications using HTML, CSS, and JavaScript following mobile-first design principles.

SELECTED PROJECTS

Distributed Observability Pipeline (Datadog-style)

github.com/KevinKattakayam/datadog

Go, Rust, Apache Kafka, ClickHouse, Prometheus, Grafana, Kubernetes, Terraform

- Built an end-to-end metrics pipeline: a Go ingestor (Gin HTTP, with gRPC/protobuf definitions) publishing validated metrics to 6-partition Kafka topics with a dead-letter queue, consumed by a Rust/tokio processor running EWMA and rolling Z-score anomaly detection.
- Ensured at-least-once delivery with no silent data loss by committing Kafka offsets only after successful ClickHouse writes, and protected the consumer loop with a circuit breaker (opens after 5 failures, exponential backoff to 60s) so backend degradation bounds lag instead of stalling ingestion. Storage is two-tier: Prometheus (15-day hot) and ClickHouse MergeTree (monthly partitions, 1-year TTL, hourly rollups).
- Built the operational layer: per-tenant token-bucket rate limiting, W3C Trace Context propagation to Tempo and Jaeger, multi-window SLO burn-rate alerts on the Google SRE model, a 12-component Helm chart with HPAs and PDBs, and a Terraform blueprint for AWS EKS. Backed by 53 unit tests (41 Go, 12 Rust), 25 integration tests, and a k6 load-test script for the 14-container stack.

PharmaTrace – Pharmaceutical Verification Platform

github.com/KevinKattakayam/pharma_trace

Python, FastAPI, LangGraph, Groq/Llama 3.3 70B, Whisper, React 19 PWA, Supabase PostGIS

- Built a 6-agent LangGraph verification pipeline (barcode, FDA lookup, recall, interaction, safety, report) over OpenFDA, RxNav, and Indian CDSCO registries, parallelizing agent lookups with `asyncio.gather` to cut end-to-end latency. Replaced naive report counts with NLM RxNorm clinical mapping, FAERS signal strength, and severity-sorted warnings.
- Enforced clinical safety with strict Pydantic schemas and a field-validator “jargon firewall” structurally blocking hallucinated medical terminology, and injected patient age, weight, and renal function into prompts to trigger dosage warnings.
- Shipped an offline-first PWA: in-browser Tesseract OCR for expiry extraction, IndexedDB caching with a Background Sync outbox, and Whisper-based vernacular voice in Malayalam, Hindi, Tamil, and English; logged every verification to a SHA-256 hash-chained audit chain; ships with Railway and Vercel deployment configs.

DevSync – Collaborative Developer Workspace

github.com/KevinKattakayam/devsync

Node.js, Express, TypeScript, Next.js, React, PostgreSQL, Prisma, Socket.io

- Built a real-time collaboration platform with JWT auth, role-based access control, Socket.io live co-editing, and Kanban task management.
- Designed the PostgreSQL schema and data access layer with Prisma, exposing a REST API with Zod validation, centralized error handling, and Jest/Supertest integration suites across auth, RBAC, and task workflows.

PatentDoc Co-Pilot – IPO-Compliant Patent Drafting

github.com/KevinKattakayam/Patent_Doc_Copilot

Python, CrewAI, LLMs, OpenRouter, Streamlit, Docker

- Built a one-click generator producing all 9 IPO-compliant patent sections with auto-generated block, flow, and sequence diagrams.
- Orchestrated a 5-agent CrewAI verification and prior-art pipeline covering claim writing, drawing description, format validation, and quality review, exporting a full filing package (Forms 1, 3, 5) as PDF and DOCX from a containerized Streamlit app.

Schamroth Window Detection – Digital Clubbing Screening

github.com/KevinKattakayam/schamroth

Python, MediaPipe, OpenCV, CLAHE, Streamlit (basis of the IEEE publication)

- Built a screening pipeline for digital clubbing using MediaPipe hand-landmark localization of opposing index fingertips, with a center-based fallback for close-up frames and pre-processing image-quality gates.
- Designed a gap-presence score from bright/dark pixel ratios, contrast, and symmetry, mapped to 5 clinical risk levels with quality-weighted confidence; evaluated with accuracy, sensitivity, specificity, F1, and a confusion matrix on a 65-image test set.

V-JEPA Video Representation Learning

github.com/KevinKattakayam/finetunevjepa

PyTorch, V-JEPA, CUDA

- Adapted pretrained V-JEPA self-supervised representations for downstream video classification using attentive probes on UCF101.
- Ran experiments over temporal sampling strategies, probe configurations, and batch sizing under constrained GPU memory.

CollegeFind – College Discovery Platform

github.com/KevinKattakayam/collegefind

Next.js 16, TypeScript, Tailwind CSS, NextAuth, Prisma, Neon PostgreSQL | Live: college-discovery-zeta.vercel.app

- Shipped a live platform covering 200+ colleges with debounced search, multi-facet filters, shareable URL state, and skeleton loaders.
- Built side-by-side comparison across 11+ metrics, an exam-rank admission predictor, saved comparisons, and a community Q&A thread with optimistic UI, deployed as a Vercel frontend proxying to a Render backend over serverless Neon Postgres.

NexusBank – Online Banking Simulation

github.com/KevinKattakayam/online_bank_simulation

Node.js, Express 5, MongoDB, Mongoose, bcrypt, express-session

- Built a banking API with bcrypt-hashed PIN authentication, MongoDB-backed sessions, balance-validated transfers, beneficiaries, and compound-indexed paginated history, plus rule-based fraud detection (amount, velocity, odd-hours, daily-limit) with a weighted 0–100 risk score and aggregation-pipeline spend analytics.

ADDITIONAL PROJECTS

- **PDF OCR & Structure Extraction Pipeline** (github.com/KevinKattakayam/pdf-ocr) – Python, Pydantic. Validation, OCR, structure extraction with fallbacks, table normalization, canonical JSON/Markdown, RAG chunking, batch CLI.
- **Invoice OCR to Excel** (github.com/KevinKattakayam/bill_excel) – PaddleOCR PP-Structure, OpenPyXL. Extracts invoice fields and line-item tables from images/PDFs, reconstructs rows/columns, exports Excel and JSON.
- **AI Research Co-Scientist** (LangChain, ChromaDB, team project) – GenAI module for literature review and hypothesis generation.
- **Research Paper Navigator** (Sentence Transformers, FAISS, Ollama) – semantic paper search with local LLM summaries.
- **Karunya Admission Guide Bot** (LangChain, FAISS) – RAG chatbot answering campus and admission queries from institutional documents.
- **GhostWorker** (FastAPI, LangGraph, Next.js) – multi-agent bias detection and legal appeal generation across 5+ jurisdictions.

TECHNICAL SKILLS

Programming Languages: Python, Go, Rust, TypeScript, JavaScript, Java, SQL

AI & Machine Learning: Large Language Models, Agentic AI, Multi-Agent Orchestration, Retrieval-Augmented Generation (RAG), Prompt Engineering, NLP, Computer Vision, Self-Supervised Learning

AI Frameworks & Libraries: LangGraph, LangChain, CrewAI, PyTorch, Hugging Face Transformers, MediaPipe, OpenCV, PaddleOCR, Tesseract, Sentence Transformers, LLM APIs, Pandas, NumPy

Backend & Web: FastAPI, Node.js, Express, Gin, gRPC, REST APIs, React 19, Next.js (App Router), Tailwind CSS, Prisma, Mongoose, Socket.io, NextAuth, JWT, RBAC, Streamlit, PWA/Workbox

Distributed Systems: Apache Kafka (KRaft), Avro & Schema Registry, Kafka Connect, at-least-once delivery, circuit breakers, per-tenant rate limiting, multi-tenancy

Databases & Stores: PostgreSQL, ClickHouse, MongoDB, MySQL, Redis, Neon, Supabase/PostGIS, ChromaDB, FAISS, Snowflake

Infrastructure & Observability: Docker, Kubernetes, Helm, Terraform, Prometheus, Grafana, OpenTelemetry, Jaeger, CI/CD, Cloudflare, Vercel, Azure, CUDA

Testing, Tooling & Foundations: Jest, Supertest, pytest, k6, Zod, Pydantic, Git, Postman, Jupyter; Data Structures & Algorithms, OOD, System Design, Concurrency, SLOs & Error Budgets

PUBLICATIONS & ACHIEVEMENTS

- **IEEE Publication (2026):** “Early Lung Cancer and Heart Disease Prediction from Clubbed Fingers Using Machine Vision”. ieeexplore.ieee.org/document/11469562
- **Completed Amazon ML Summer School 2026** (Amazon)
- **McKinsey Forward** (McKinsey & Company)

CERTIFICATIONS

Professional Certifications: Microsoft Certified: Azure AI Engineer Associate (AI-102) · Microsoft Certified: SQL AI Developer Associate (DP-800) · SnowPro Associate: Platform (Snowflake) · Agentforce Specialist (Salesforce) · Oracle Agentic AI Foundations Associate · Oracle OCI AI Foundations Associate · Redis Associate Developer · Certified LLM Security Professional (CLLMSP) · Google Cybersecurity Professional · Python Web Developer (ICT Academy – Infosys)

Courses & Credentials: Google AI Essentials · Python for Data Science (IBM) · Deep Learning Fundamentals (Lightning AI) · Data Science for Engineers (NPTEL) · Google Cloud Computing Foundations · Responsive Web Design (freeCodeCamp) · Copado AI Certified · Microsoft Applied Skills · Google Cloud Skill Badges